MATH 526 Lab 8 Fall 2006

Part I

In this part we learn to use the MATLAB commands **norm** and **cond**.

MATLAB has a built-in command **norm** for the computation of vector norms and matrix norms. Create

$$\mathbf{x} = \begin{bmatrix} -2 \\ 3 \\ 5 \\ -1 \end{bmatrix} \qquad \mathbf{A} = \begin{bmatrix} 1 & 0 & -2 & 5 \\ -3 & 2 & 1 & -4 \\ 5 & -6 & -1 & 0 \\ 2 & 5 & 0 & 12 \end{bmatrix}$$

Try the following commands:

>>**norm(x,1)**

This computes the 1-norm of $\mathbf{x}$.

>>**norm(x,2)**

This computes the 2-norm of $\mathbf{x}$.

>>**norm(x)**

The default norm for a vector is the 2-norm.

>>**norm(x,inf)**

This computes the ∞-norm of $\mathbf{x}$.

>>**norm(A, 1)**

This computes the 1-norm of $\mathbf{A}$.

>>**norm(A, 2)**

This computes the 2-norm of $\mathbf{A}$.

>>**norm(A)**

The default norm for a matrix is the 2-norm.

>>**norm(A, inf)**

This computes the ∞-norm of $\mathbf{A}$.

The condition number $\kappa(\mathbf{A})$ of a nonsingular $n \times n$ matrix $\mathbf{A}$ is defined by the formula

$$\kappa(\mathbf{A}) = \|\mathbf{A}\| \cdot \|\mathbf{A}^{-1}\|\,.$$

The three commonly used condition numbers are

$$\kappa_1(\mathbf{A}) = \|\mathbf{A}\|_1 \cdot \|\mathbf{A}^{-1}\|_1\,, \quad \kappa_2(\mathbf{A}) = \|\mathbf{A}\|_2 \cdot \|\mathbf{A}^{-1}\|_2 \quad \text{and} \quad \kappa_\infty(\mathbf{A}) = \|\mathbf{A}\|_\infty \cdot \|\mathbf{A}^{-1}\|_\infty\,.$$

MATLAB has a built-in command **cond** for the computation of the condition number of a matrix.

\>>**cond(A)**

The default condition number for a matrix is the condition number $\kappa_2(\mathbf{A})$.

Of course you can also compute $\kappa_2(\mathbf{A})$ by the following commands:

\>>**norm(A)*norm(inv(A))**

The MATLAB command **inv(A)** computes the inverse of the matrix **A**.

You can also compute $\kappa_1(\mathbf{A})$ and $\kappa_\infty(\mathbf{A})$:

\>>**cond(A,1)**

This computes the condition number $\kappa_1(\mathbf{A})$.

\>>**cond(A, inf)**

This computes the condition number $\kappa_\infty(\mathbf{A})$.

Clear all variables before you work on Part II